MUHSIN AHMED

Full-Stack Engineer & Data Scientist

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PROFILE

Full-stack engineer with a BSc in Data Science & Analytics, building a production e-government platform (Go, React, TypeScript, PostgreSQL) that has processed **53,000+ applications** and **20,000+ payments**. I deliver features end to end, from database and API design to user interfaces, document generation and payment integration, and bring machine learning, predictive analytics and data visualization skills (Python, Power BI) to the products I build.

PROFESSIONAL EXPERIENCE

Full-Stack Software Developer

One Stop Development · Remote/Hybrid · 2024 – Present

- Build end-to-end features for a production e-government platform that digitizes land administration, building permits, property tax, civil registration and revenue collection for a regional government; **53,000+ applications** since January 2026, over 21,000 of them in October 2026 alone.
- Develop the Go REST API (Gin, PostgreSQL, NATS) and the React/Remix + TypeScript frontends (admin portal, citizen portal, mobile) in an Nx monorepo.
- Built a **no-code workflow composer** that lets administrators design, version and publish multi-step approval routes with role-based steps and automated tasks.
- Built **official document generation** for title deeds, building permits, and birth, marriage and death certificates in Somali, Amharic and English, with QR verification and digital signatures; **16,000+ documents** issued since August 2026, including 2,500+ land certificates and building permits.
- Integrated **eBirr mobile-money payments**, cash receipts and SMS payment reminders, with **20,000+ payments** completed through the platform.
- Built **analytics and reporting on PostgreSQL**: an executive dashboard backed by 17 SQL views and 10 filterable report templates, property-tax and delinquency reports reconciled to the cent, officer-productivity reports and Excel exports.
- Cut application search from **10+ minute timeouts to under 2 seconds** by rewriting the query; added role-based access control, login auditing and an admin screen lock.
- Work test-first and specification-driven, with code review and GitLab CI releases to staging and production.

Data Scientist & Frontend Developer

Faras Cabs · Nairobi, Kenya · 2023 – 2024

- Designed and built interactive **Power BI dashboards** tracking ride patterns, revenue trends and operational efficiency for management.
- Cleaned and analyzed operational data and produced regular reports to support business decisions.
- Built responsive React interfaces and contributed to backend development, including database integrations.
- Worked with developers and IT staff to support the day-to-day operation of production systems.

SELECTED PROJECTS

Nairobi Ride-Hailing Surge Pricing Prediction & Spatial Mapping

BSc capstone · Python, XGBoost, Optuna, GeoPandas, Folium, Streamlit · github.com/Muhsin48ahmed/nairobi-surge-analysis

- Predicted whether a trip would be surge-priced from **3.7M Nairobi ride-hailing trips** merged with hourly weather. Compared six classifiers with Optuna tuning on time-ordered cross-validation; a calibrated XGBoost model reached **ROC-AUC 0.78 on a sealed test set**.
- Guarded against leakage (chronological splits, excluded leaking features) and reported honestly that surge **size** is not predictable from trip-level data.
- Mapped where and when surge concentrates (evening peaks 17:00–21:00, rain amplification, a CBD hotspot cluster) with GIS joins, Folium heatmaps and K-means, and triangulated the findings with **35 driver, passenger and expert interviews**.
- Delivered a three-page Streamlit decision-support dashboard (overview, hotspot map, predict), backed by pytest tests and reproducibility documentation.

Unified Public Health Information System (UPHIS)
Mombasa County · Role: data collection & analysis

Handled data collection and analysis in a cross-functional team that built and deployed a unified public health information system, consolidating multiple health data sources into one platform. Improved access to data for county public-health decisions and supported the move toward universal health coverage.

Customer Churn Prediction
Team project · Python, scikit-learn, XGBoost, SMOTE

Built a churn model on 3,333 customer records with 21 features, using SMOTE for class imbalance and comparing logistic regression, decision tree, random forest and XGBoost. The tuned XGBoost model reached **0.73 recall and 0.73 F1** on churning customers (92.4% accuracy against an 86% majority-class baseline). Translated the results into customer-retention recommendations.

Stream254 — Movie Recommendation System
Python, Surprise, pandas

Built collaborative-filtering recommenders on 100,000+ MovieLens ratings, comparing KNN, SVD and NMF models with RMSE and MAE. SVD cut test RMSE to **0.79 from a 0.94 KNN baseline**. Addressed cold-start and data-sparsity challenges.

EDUCATION

BSc Data Science & Analytics
United States International University–Africa, Nairobi · Conferred September 2026

Capstone: Nairobi ride-hailing surge pricing prediction & spatial mapping (see Projects).

Relevant coursework: Machine Learning, Data Mining & Analysis, Big Data Analytics, Statistical Modelling, Data Warehousing, Natural Language Processing, Mathematical Finance.

Data Science Core Program
Moringa School, Nairobi · 2024

Intensive program in applied data analysis, data visualization and machine learning.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL, R
Engineering: React, Remix, Gin, REST APIs, NATS, Nx monorepo, Tailwind CSS, HTML/CSS
Data & ML: pandas, NumPy, scikit-learn, XGBoost, TensorFlow, statistical modelling, NLP
Databases & BI: PostgreSQL, ClickHouse, MySQL, MongoDB, Power BI, Matplotlib, Seaborn
Tools: Git, GitHub, GitLab CI, Jupyter Notebook, Insomnia, ClickUp

AWARDS & LEADERSHIP

Certificate of Honour — E-Government Platform Implementation
City Administration Revenue Office · September 2026

Presented to the platform's implementation team for its key role in rolling out the new system.

President's Award — Gold Standard
The Duke of Edinburgh's International Award · December 2020

Highest level of the award, presented by the President of Kenya for sustained community service, skills development, physical recreation and expedition.

Coordinator, Kenya-USA Tech Forum (KUSAT) Hackathon
Managed event planning, sponsor engagement and participant coordination for a hackathon connecting the Kenyan and US tech communities.

LANGUAGES

English · Swahili · Somali · Arabic

References available on request.